

# Pooya Eini, MD

Physician-Researcher | Cardiovascular Research & Clinical Data Science

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## RESEARCH PROFILE

Physician and clinical researcher focused on cardiovascular imaging, clinical prediction and evidence synthesis. My contributions include statistical analysis and full manuscript drafting for original studies, alongside leadership of systematic reviews from literature searching through quantitative synthesis and writing. Seeking doctoral training that builds on this clinical and analytical foundation, including opportunities linking cardiovascular phenotypes with molecular mechanisms and biomarkers.

## EDUCATION & LANGUAGE

**Doctor of Medicine (MD)** | Shahid Beheshti University of Medical Sciences, Tehran | 2017-2024

**GPA: 18.35/20** (original Iranian university grading scale).

**English: IELTS overall band 7.0** (expired; historical result).

## RESEARCH EXPERIENCE & AFFILIATIONS

### Clinical researcher

Shaheed Rajaie Cardiovascular Medical and Research Center, Tehran | 2025-present

Infectious Disease Research Center, Hamadan University of Medical Sciences, Hamadan, Iran | 2023-present

- **Original studies:** performed statistical analyses and wrote complete manuscript drafts, including methods, results and interpretation; incorporated co-author feedback and revisions.
- **Systematic reviews and meta-analyses:** led literature searches, screening, data extraction, critical appraisal, statistical synthesis and full manuscript drafting.

## EDITORIAL SERVICE

**Editorial Review Board** | Sage Open | 2026-present

## APPLIED RESEARCH SKILLS

- **Clinical data analysis:** regression, machine-learning model development and evaluation, internal validation, discrimination, calibration and decision-curve analysis.
- **Research programming:** applied use of R and Python; statistical analysis and evidence synthesis using R, Stata and SPSS. Training developed through online coursework, self-study and AI-assisted learning and coding.
- **Evidence synthesis:** systematic reviews of diagnostic tests, prediction models and biomarkers; critical interpretation of methodological limitations and clinical relevance.
- **Scientific communication:** complete manuscript preparation, revision and explanation of quantitative findings in a clinical context.

## SELECTED ONLINE COURSEWORK

**Imperial College London:** Linear Regression in R for Public Health; Introduction to Statistics and Data Analysis in Public Health; Immunology: The Immune System and Its Failures.

**Johns Hopkins University:** Computing for Cancer Informatics; Cancer Biology. **Yale University:** Introduction to Breast Cancer.

## SELECTED COLLABORATIVE PROJECTS

### **EACVI-MMVD (Multiple and Mixed Valvular Disease Study) | European Association of Cardiovascular Imaging**

Collaborator in a multinational study involving more than 30 centres worldwide; contributed to multicentre data collection for the standardized evaluation of multiple and mixed valvular heart disease.

### **DataTypeHub | Project Pars Peace | GitHub repository**

Contributed to data gathering for an open-source, centralized resource presenting diverse medical data modalities and practical best practices for streamlined data workflows.

## SELECTED ORIGINAL RESEARCH

Selected publications with analytical and manuscript-writing contributions.

Eini P, Serpoush H, Rezayee M. **Explainable machine learning and unsupervised anatomical phenotyping of mitral valve prolapse in patients with severe mitral regurgitation.** BMC Cardiovascular Disorders. 2026. doi:10.1186/s12872-026-06331-5

**First author.** Performed the analysis and wrote the full manuscript. Analyzed a public cohort of 979 patients using supervised learning and anatomical clustering; assessed internal validation, calibration and clinical utility.

Zahedi M\*, Eini P\*, Alemzadeh-Ansari MJ, Motevalli M, Fallah Najmabadi A, Houshmand G. **Left main coronary artery stenosis assessment on coronary CT angiography.** European Journal of Radiology Open. 2026;17:100803. doi:10.1016/j.ejro.2026.100803

**\*Equal contribution / co-first author.** Performed statistical analysis and wrote the full manuscript evaluating CCTA-based assessment of left main coronary stenosis.

Ariannia R, Badkoubi A, Shirini D, et al. **Impact of lead placement site on right ventricular function: a comparative echocardiographic analysis of apical versus septal pacing in permanent pacemaker recipients.** American Journal of Cardiovascular Disease. 2025;15(5):308-318. doi:10.62347/AMNY1624

**Co-author.** Performed statistical analysis and wrote the full manuscript comparing right-ventricular function by pacing-lead placement.

Mostafazadeh B, Hosseini SM, Shadnia S, et al. **Machine Learning-Based Prognostic Prediction Models in Calcium Channel Blockers Poisoning.** Archives of Academic Emergency Medicine. 2025;13(1):e79. doi:10.22037/aaem.v13i1.2804

**Co-author.** Performed the analysis and wrote the full manuscript on prognostic modeling in acute poisoning.

Meshkini M, Hosseini SM, Erfan Talab Evini P, et al. **Predicting the Risk of Opioid-induced Respiratory Depression Using ChatGPT-4o and Machine Learning Techniques.** Archives of Academic Emergency Medicine. 2025;13(1):e85. doi:10.22037/aaem.v13i1.2832

**Co-author.** Performed the analysis and wrote the full manuscript evaluating prediction of opioid-induced respiratory depression.

Eini P, Serpoush H, Rezayee M, Tremblay J. **Machine-learning prediction of cardiovascular diagnoses from pediatric ECG reports in 11 643 children.** BMC Pediatrics. 2026. doi:10.1186/s12887-026-07732-3

**First author.** Performed model development and analysis and wrote the full manuscript using structured pediatric ECG reports from a public dataset.

Eini P, Ghaffari Jolfayi A, Firouzi A, et al. **CT mitral valve calcium score and immediate outcomes after percutaneous transvenous mitral commissurotomy.** BMC Medical Imaging. 2026. doi:10.1186/s12880-026-02843-w

**Co-first author with Dr Mahdi Zahedi.** Dr Golnaz Houshmand is the last and corresponding author. Performed statistical analysis and wrote the full manuscript.

## SELECTED CASE REPORT

Eini P, Mahdavi M, Ghaemmaghami Z, Pouraliakbar H, Houshmand G. **Heart failure in a 10-year-old girl: incidental discovery of adrenal adenoma by cardiac MRI—case report.** European Heart Journal - Case Reports. 2026;10(3):ytag110. doi:10.1093/ehjcr/ytag110

## SELECTED FIRST-AUTHOR SYSTEMATIC REVIEWS

Selected cardiovascular and cardiopulmonary evidence-synthesis publications.

Eini P, Eini P, Serpoush H, et al. **Diagnostic accuracy of machine-learning models for detecting pulmonary embolism in patients with pneumonia: a systematic review and meta-analysis.** BMC Pulmonary Medicine. 2026. doi:10.1186/s12890-026-04583-7

Eini P, Serpoush H, Rezayee M, Kassulke M. **Artificial Intelligence-Based Analysis of Coronary Atherosclerotic Plaque on Intravascular Ultrasound: A Systematic Review and Meta-Analysis.** Clinical Cardiology. 2026;49(7):e70423. doi:10.1002/clc.70423

Eini P, Serpoush H, Rezayee M, Kassulke M. **Diagnostic Performance of Artificial Intelligence in Evaluating Tricuspid Regurgitation: A Systematic Review and Meta-Analysis.** Clinical Cardiology. 2026;49(6):e70366. doi:10.1002/clc.70366

Eini P, Rezayee M, Tremblay J. **Automated assessment of right heart function by artificial intelligence: A systematic review and meta-analysis.** European Journal of Radiology Open. 2026;16:100713. doi:10.1016/j.ejro.2025.100713

Eini P, Houshmand G, Rezayee M, Kassulke M. **Accuracy of machine learning models for mitral regurgitation severity assessment: a systematic review and meta-analysis.** International Journal of Cardiology Cardiovascular Risk and Prevention. 2026:200642. doi:10.1016/j.ijcrp.2026.200642

Eini P, Rezayee M, Kassulke M. **Machine Learning and Regression-Based models for Prediction of Postoperative Atrial Fibrillation Following Coronary Artery Bypass Grafting: A Systematic Review and Meta-Analysis.** International Journal of Cardiology Cardiovascular Risk and Prevention. 2026:200619. doi:10.1016/j.ijcrp.2026.200619

Eini P, Eini P, Rezayee M, Tremblay J. **MicroRNA expression profiles in abdominal aortic aneurysms: A systematic review of potential diagnostic and prognostic biomarkers.** International Journal of Cardiology Cardiovascular Risk and Prevention. 2026:200651. doi:10.1016/j.ijcrp.2026.200651

Eini P, Eini P, Serpoush H, Rezayee M. **Diagnostic performance of machine learning algorithms for predicting heart failure in diabetic patients: a systematic review and meta-analysis.** Endocrinology, Diabetes & Metabolism. 2025;8(5):e70111. doi:10.1002/edm2.70111

## SELECTED MOLECULAR & TRANSLATIONAL REVIEWS

Eini P [Peyman], Eini P [Pooya], Amini NB, Farzan P, Etemady K. **Regulatory pathways of long non-coding RNAs and microRNAs in sepsis-induced cardiomyopathy.** Discover Medicine. 2025;2:289. doi:10.1007/s44337-025-00530-3

Eini P [Peyman], Eini P [Pooya], Pourhemmati S, et al. **Monocytes in HIV associated atherosclerosis: A review of pathogenesis and clinical implications.** Caspian Journal of Internal Medicine. 2026;17(1):1-13. doi:10.22088/cjim.17.1.1

## ACCEPTED / IN PRESS

**Prevalence, risk factors, and outcomes of infective endocarditis in people living with HIV: a systematic review and meta-analysis.** Caspian Journal of Internal Medicine; in press.

**The gut-heart axis in acute coronary syndromes: a systematic review of gut microbiota, metabolites, and systemic markers.** Journal of Lipid and Atherosclerosis; accepted August 2026.

Full publication record: [Google Scholar](#)